

Week 7 Pre-Class Study Guide

Topic: Deep Learning Best Practices **Estimated Time:** 25 minutes **Due:** Before Week 7 live session

Overview

This week you'll learn production-ready deep learning practices: diagnosing overfitting, applying regularization, and saving models. Week 7 builds directly on Week 6's neural network foundations.

Learning Objectives

By the end of Week 7, you will:

1. Diagnose overfitting by plotting training/validation curves
 2. Implement three-way data split (train/validation/test)
 3. Apply Dropout regularization to prevent overfitting
 4. Use EarlyStopping callback to stop training automatically
 5. Save and load complete Keras models
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Review: Week 6 Concepts

Before Week 7, make sure you understand these Week 6 concepts:

Keras Sequential API

```
from keras import models, layers

model = models.Sequential([
    layers.Dense(128, activation='relu'),
    layers.Dense(10, activation='softmax')
])
```

Compile, Fit, Predict Pattern

```
# Compile
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# Fit (train)
history = model.fit(X_train, y_train, epochs=10)

# Predict
predictions = model.predict(X_test)
```

Training History

```
# Plot training curves
import matplotlib.pyplot as plt
plt.plot(history.history['loss'])
plt.plot(history.history['accuracy'])
```

Self-Check: Can you build and train a simple neural network on MNIST from memory?

New Concepts for Week 7

1. Overfitting (The Problem We're Solving)

Definition: Model memorizes training data instead of learning general patterns.

Symptoms:

- Training accuracy: 95%
- Validation accuracy: 85%
- Large gap between train and validation performance

Analogy: Student who memorizes exact practice problems but can't solve new problems.

2. Three-Way Data Split (Our Diagnostic Tool)

Week 6: Train/Test split (2 sets) **Week 7:** Train/Validation/Test split (3 sets)

Set	Purpose	When Used
Training	Update model weights	During <code>model.fit()</code>
Validation	Monitor overfitting	During <code>model.fit()</code> (via <code>validation_data</code>)
Test	Final evaluation	After training complete (via <code>model.evaluate()</code>)

Analogy:

- Training set = Textbook (you learn from it)
- Validation set = Practice test (check progress while studying)
- Test set = Final exam (evaluate once at end)

Code:

```
from sklearn.model_selection import train_test_split

X_train, X_val, y_train, y_val = train_test_split(
    X_train_full, y_train_full, test_size=0.2, random_state=42
)
```

3. Training Curves (Diagnosing Overfitting)

Healthy Training:

- Training loss: ↓ decreasing
- Validation loss: ↓ decreasing

Overfitting:

- Training loss: ↓ decreasing
- Validation loss: ↑ increasing

Key Insight: If training and validation curves diverge, you're overfitting!

4. Dropout (Our First Fix)

What it does: Randomly drops 30% of neurons during training

Why it helps: Forces network to learn robust features, not rely on specific neurons

Where to place: After Dense layers (not after output layer)

Code:

```
from keras import layers

model = models.Sequential([
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.3), # Drop 30% of neurons
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.3),
    layers.Dense(10, activation='softmax') # NO Dropout here!
])
```

Common dropout rates: 0.2 (light), 0.3 (default), 0.5 (heavy)

5. EarlyStopping (Our Second Fix)

What it does: Automatically stops training when validation loss stops improving

Why it helps: Prevents overfitting by not training too long

Code:

```
from keras.callbacks import EarlyStopping

early_stop = EarlyStopping(
    monitor='val_loss', # Watch validation loss
    patience=3, # Wait 3 epochs before stopping
    restore_best_weights=True, # Revert to best epoch
    verbose=1
)

history = model.fit(
    X_train, y_train,
    validation_data=(X_val, y_val), # Required!
    epochs=50,
    callbacks=[early_stop]
)
```

Key parameter: patience=3 means “if validation loss doesn’t improve for 3 consecutive epochs, stop”

6. Model Saving & Loading

Save complete model:

```
model.save('my_model.keras')
```

Load model:

```
from keras.models import load_model
loaded_model = load_model('my_model.keras')
```

What's saved: Architecture + weights + optimizer state (everything!)

Required Pre-Class Viewing

Video 1: StatQuest - Bias and Variance (6 min)

URL: <https://www.youtube.com/watch?v=EuBBz3bI-aA>

What you'll learn:

- Bias = underfitting (model too simple)
- Variance = overfitting (model too complex)
- Visual intuition for overfitting

Key Takeaway: Overfitting happens when model is too complex for the data.

Reading: Keras Training Guide - Callbacks Section (15 min)

URL: https://keras.io/guides/training_with_built_in_methods/

What to focus on:

- How to use `validation_data` in `model.fit()`
- `EarlyStopping` callback examples
- `ModelCheckpoint` callback basics

Don't worry about: Custom callbacks, advanced training loops

Fashion-MNIST Dataset Preview

Week 7 uses **Fashion-MNIST** instead of MNIST:

Similarities to MNIST:

- 28×28 grayscale images
- 60,000 training, 10,000 test
- 10 classes
- Same loading API

Differences:

- Clothing items (not digits)
- Slightly harder to classify
- More realistic challenge

Classes:

0. T-shirt/top
1. Trouser
2. Pullover
3. Dress
4. Coat
5. Sandal
6. Shirt
7. Sneaker
8. Bag

9. Ankle boot

Why Fashion-MNIST? Harder than MNIST → overfitting more likely → better for learning regularization!

Pre-Class Practice (Optional)

Notebook: `week7_preclass_practice.ipynb` (in this folder)

What it covers: - Load MNIST (refresher from Week 6) - Build simple neural network - Plot training history - Practice train/test split

Estimated time: 15-20 minutes

Scaffolding: 70% provided (focus on reviewing concepts)

Self-Check Questions

Before Week 7, can you answer these?

1. What's the difference between training and validation sets?

Answer

Training set: used to update weights. Validation set: used to monitor overfitting (not used for weight updates).

2. How do you recognize overfitting from training curves?

Answer

Training loss decreasing, but validation loss increasing (or flat). Large gap between training and validation accuracy.

3. Where should you place Dropout layers?

Answer

After Dense layers, before the next layer. NOT after the output layer.

4. What does `patience=5` mean in `EarlyStopping`?

Answer

If validation loss doesn't improve for 5 consecutive epochs, stop training.

5. What file extension should you use to save Keras 3.x models?

Answer

.keras (not .h5, which is legacy format)

Common Confusions (Preview)

“Do I turn off Dropout during prediction?”

→ No! Keras does this automatically.

“Why is my validation loss higher from epoch 1?”

→ This is normal with Dropout. Look for divergence over epochs.

“What if my training stops at epoch 10 when I set epochs=50?”

→ That’s EarlyStopping working! It stopped early because validation loss stopped improving.

What to Bring to Live Session

1. **Completed pre-class viewing** (videos + reading)
 2. **Questions** about overfitting, Dropout, or callbacks
 3. **Laptop** with Jupyter and Keras 3.x installed
 4. **Refreshed memory** of Week 6 Keras basics
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Week 7 Roadmap

Live Session (3 hours):

1. Week 6 recap (15 min)
2. Three-way data split (30 min)
3. Overfitting deep dive (30 min)
4. Break (15 min)
5. Regularization techniques (30 min)
6. Live coding: Fashion-MNIST complete pipeline (35 min)
7. Pair programming: “Regularization Detective” (20 min)
8. Wrap-up (5 min)

Post-Class:

- Build Fashion-MNIST classifier from scratch
 - Experiment with dropout rates
 - Practice using callbacks
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Key Vocabulary

Term	Simple Definition
Overfitting	Model memorizes training data, performs poorly on new data
Validation set	Data used to monitor overfitting during training
Dropout	Randomly deactivate neurons during training
Callback	Function that runs at specific points during training
Patience	How many epochs to wait before EarlyStopping triggers
restore_best_weights	Revert to best epoch when stopping

Next Steps

1. Watch StatQuest video (6 min)
2. Read Keras training guide - callbacks section (15 min)
3. Review this study guide (you’re doing it now!)
4. (Optional) Complete pre-class practice notebook
5. (Optional) Download Fashion-MNIST dataset

Questions Before Class?

If anything is unclear, don't stress! The live session will bring it all together.

See you in Week 7!

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